

reasonably matching the performance of comparable GPU-based open-weight models. This makes it possible to run LLMs on commodity hardware. Bitnet comprises a model family (like BitNet b1.58 2B4T, a 2-billion-parameter LLM trained on 4 trillion tokens) and an open source inference framework (bitnet.cpp) optimised for these 1-bit models. According to reports, BitNet b1.58 2B4T achieves performance on par with top open-weight models of similar size, while cutting memory and energy by approximately 70%. In November 2024, Microsoft Research introduced BitNet a4.8, enabling 4-bit activations for 1-bit LLMs.

Kompact AI. Kompact AI, developed by Ziroh Labs in collaboration with IIT Madras, runs AI models efficiently on standard CPUs, eliminating the need for expensive GPUs for inference, fine-tuning, and even light training. At its core is the ICAN, or Common AI-Language Runtime, which optimises model execution by employing techniques like quantisation, pruning, and distillation to reduce computational load. It intelligently leverages CPU strengths, such as multi-core processing and SIMD instructions, to handle workloads. This helps deliver up to 3x the performance of traditional CPU execution in frameworks like TensorFlow or PyTorch for optimised models.

Prof. V. Kamakoti, Director of IIT Madras, explains in a press release that the most important aspect of this product is that it can be scaled depending upon the need. “For example, if you have a language model that you want to build for, say, 500 million parameters, you can purchase a conventional CPU hardware of around ₹150,000 and do the inferencing part of it. Of course, the training must be done in a GPU environment, but the inferencing can happen here. Similarly, if you want to do 7 billion parameters, probably

a conventional server of ₹800,000-1,000,000 will be sufficient. Like this, as we scale up from smaller language models, we can also scale up the hardware, and this is one of the most important contributions of Kompact AI.”

He remarks that solutions like Kompact AI will play a significant role in democratising AI—or bridging the AI divide, as he terms it. As an example, “Suppose the government decides to provide AI assistance for history students across classes. They can train the model at a central place, hiring GPUs. A trained model can be put in the schools using conventional servers with Kompact AI software layer and the kids can use it. So, for the inferencing part, we do not need GPUs. It is a very cost-effective solution.”

Arm. Arm has enhanced its architecture with features like scalable matrix extension (SME and SME2), which improves the performance of matrix operations for AI, ML, and high-performance computing. It has also invested heavily in extensive software libraries and tools to ensure easy optimisation, development, and deployment of AI on Arm CPUs. Kleidi is a software initiative and suite of open source libraries developed by Arm, which includes the KleidiAI library of optimised micro-kernels for AI-specific workloads, and KleidiCV, for enhancing computer vision applications. Kleidi integrates easily into AI frameworks like PyTorch, ExecuTorch, and MediaPipe. Lumex, Arm’s new compute subsystem platform integrates Armv9.3-A CPUs with SME2 for AI acceleration, a Mali GPU with Ray Tracing v2, and unified software tools to enable real-time, on-device intelligence.

MiPhi Semiconductors, Intel, and SUSE collab. Bengaluru-based MiPhi Semiconductors has collaborated with Intel and SUSE to develop a solution for training small-scale AI models (like NLP,

computer vision, and recommendation systems) on standard CPUs, without depending on expensive and resource-intensive GPUs. The strategic partnership brings together Intel’s advanced CPU architectures, SUSE’s scalable enterprise-grade Linux platform, and MiPhi’s semiconductor expertise and aiDAPTIV+ platform for LLM training and AI inferencing.

GGML and llama.cpp. GGML, or GPT-Generated Model Language, is a tensor library designed to run large AI models on commodity hardware like CPUs by optimising their architecture. The open source project llama.cpp wraps around GGML, enabling users to run models such as Llama on CPUs.

Winds of change

The AI landscape is undergoing a foundational shift, challenging the monopoly of GPU-based computing, more specifically Nvidia GPUs. While Nvidia remains the market leader with its powerful hardware and CUDA software ecosystem, technological, economic, environmental, and geopolitical forces are causing companies to look for alternatives. We see hyperscalers like Google, Amazon, Microsoft, and OpenAI developing specialised AI chips (ASICs/AI accelerators) in-house, while smaller players are exploring the possibility of running AI workloads on ubiquitous CPUs.

Simultaneously, geopolitical undercurrents are reshaping the global supply chain. The US’s export restrictions on advanced chips have compelled countries like China and India to seek self-sufficiency. India has launched ambitious initiatives to develop indigenous GPUs and foundational AI models, supported by significant public investment in computing infrastructure, and unbridled enthusiasm and talent from the private sector. The winds of change are evident, signalling a move toward a more decentralised, accessible, and sustainable AI future. **EFY**